CAP High School/CEGEP Prize Exam

April 09, 2025

9:00 - 12:00

Student Information Sheet

The following information will be used to inform students and schools of the exam results, to determine eligibility for some subsequent competitions and for statistical purposes. Only the marking code, to be assigned by the local examination committee, will be used to identify papers for marking.

Marking Code:

This box must be left empty.

PLEASE PRINT CLEARLY IN BLOCK LETTERS

Student First Name																										
Student Last Name																										
Teacher Name																										

Address: (Street number and Apt) _____

(City and Province) _____ (Postal Code) _____

Telephone: ( ) _____ Email: _____

School: _____ Grade: _____

Date of Birth: _____

Gender: _____ Citizenship: _____

If you are not a Canadian citizen, what is your Immigration Status? _____

For how many years have you studied in a Canadian school? _____

Would you prefer further correspondence in French or English? _____

By signing this page I agree that, if my score on this Exam is in the top 20 nationally or in the top 6 in my

province, my name, my school's name, and my ranking can be published.

Student's signature: _____

Sponsored by:

Canadian Association of Physicists; Canadian Physics Olympiad; Department of Physics and Astronomy, University of British Columbia.

2025 Canadian Association of Physicists Highschool/Cegep Prize Exam

This is a three-hour exam. National ranking and prizes will be based on students' performance on sections A and B of the exam. Performance on questions in section A will be used to determine whose written work in section B will be marked for prize consideration by the CAP Exam National Committee. Section A consists of 25 multiple-choice questions. The questions in section B span a range of difficulty, and may require graphing and/or measurement on the graph. Be careful to gather as many of the easier marks as possible before venturing into more difficult territory. When you are unable to solve any part of a question, you may assume a likely answer to that part and attempt the rest of the question anyway.

Non-programmable calculators may be used. Answer the multiple-choice questions ***on the answer sheet*** provided. Most importantly: write your solutions to the three long problems on ***three separate sheets*** as they will be marked by people in different parts of Canada. Good luck.

Notice: Full marks will be given to students who provide full correct solutions to problems in Section B. Partial marks will be given for partial solutions. There are no penalties for incorrect answers. The questions are not of equal difficulty. Remember that we are challenging the best physics students in Canada; it is possible that even the best papers may not achieve an overall score of 80%. This Exam is meant to be challenging!

Data

Speed of light $c = 3.00 \times 10^8$ m/s
Speed of sound in air = 343 m/s
Gravitational constant $G = 6.67 \times 10^{-11}$ N·m^2/kg^2
Acceleration due to gravity $g = 9.81$ m/s^2
Standard atmospheric pressure $P_0 = 1.01 \times 10^5$Pa
Density of fresh water $\rho = 1.00 \times 10^3$ kg/m^3
Density of air at 20°C and standard pressure $\rho = 1.2$ kg/m^3
Density of ice $\rho_i = 916$ kg/m^3
Specific heat of water $C_w = 4186$ J/(kg·K)
Specific heat of ice $C_i = 2050$ J/(kg·K)
Latent heat of water $L_w = 2260$ kJ/kg
Latent heat of ice $L_i = 334$ kJ/kg
Fundamental charge $e = 1.60 \times 10^{-19}$ C
Mass of an electron $m_e = 9.11 \times 10^{-31}$ kg
Mass of a proton $m_p = 1.67 \times 10^{-27}$ kg
Planck's constant $h = 6.63 \times 10^{-34}$ J·s
1 eV $\approx 1.602\times10^{-19}$ J
Electrostatic constant $k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 N \cdot m^2/C^2$
Permittivity of free space $\epsilon_0 = 8.854\times10^{-12}$ C^2/N·m^2
Boltzmann's constant $k_B = 1.38 \times 10^{-23}$ J/K
Stefan–Boltzmann constant $\sigma = 5.67 \times 10^{-8}$ W/m^2·K^4
Astronomical Unit (approximate distance from the Sun to the Earth) 1 AU = 1.49598×10^{11} m
Radius of the Earth $R_E = 6.371 \times 10^6$ m
Radius of the Sun $R_S = 6.96 \times 10^8$ m
Mass of the Earth 5.97×10^{24} kg
Mass of the Sun 1.99×10^{30} kg
H_2 Molar mass 2.016 g/mol
O_2 Molar mass 31.998 g/mol
N_2 Molar mass 28.013 g/mol

Section A

1. LEDs red, green and blue (all of the same power) are connected in parallel to a variable voltage source. The voltage is slowly increased from 0. Which diode will light up first?

   A. Red

   B. Green

   C. Blue

   D. All will light up at the same voltage.

   Solution: A.
   Red LED will light up first as the energy of the red photons is lowest and the voltage needed to deliver sufficient energy will be lowest.

2. A sinusoidal voltage wave form with frequency of 10 Hz is delivered to a small incandescent bulb. The light coming from the bulb will be:

   A. Steady

   B. Flickering at 10 Hz

   C. Flickering at 5 Hz

   D. Flickering at 20 Hz

   Notice that we can perceive flickering only at frequency less than than 30 Hz.

   Solution: D.
   The intensity of incandescent light depends only on the magnitude of the current. Sufficient current in positive or negative direction will light it up. So it will light up twice per sinusoidal period.

3. The same sinusoidal voltage wave form with frequency of 10 Hz is delivered to an LED. The light coming from the LED will be:

   A. Steady

   B. Flickering at 10 Hz

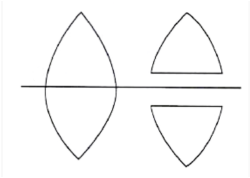
   C. Flickering at 5 Hz

   D. Flickering at 20 Hz

   Notice that we can perceive flickering only at frequency less than 30 Hz.

   Solution: B.
   10 Hz LED is a diode, so only current in one direction creates the current, which produces the light.

4. A thin lens of focal length $f = 10$ cm is cut into two equal halves. What is the new focal length of each half?



   A. 2.5 cm

   B. 5 cm

   C. 10 cm

   D. 20 cm

   E. 40 cm

   Solution: C.
   Cutting the lens in the specified manner only changes the amount of incident light that goes through the lens. It does not change the focal length.

5. When we see a half moon (first or third quarter moon), the amount of energy reaching the moon from the Sun is:

   A. about half of the energy reaching the moon from the Sun compared to when we see full moon

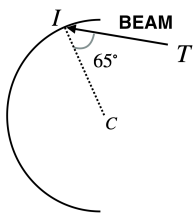
   B. about the same energy reaching the moon from the Sun compared to when we see full moon.

   C. about a quarter of the energy reaching the moon from the Sun when we see a full moon

   D. depends on the season.

   Solution: B.
   The way we see the moon does not change the way it is illuminated by the Sun.

6. The internal surface of a semi-cylinder shown in the diagram is a mirror. A narrow beam of light propagating in a plane perpendicular to the axis of the semi-cylinder hits the upper edge at point I, at an angle of 65° to the line $I - C$. Point C in the figure is located on the axis of the semi-cylinder. After how many reflections will this ray leave the semi-cylinder?



A. 1

B. 2

C. 3

D. 4

Solution: D.
The beam TI reflects at an angle of 65° to the normal CI. Thus the light ray deflects from its original direction by $2\times(90°-65°) = 50°$. Using the geometry of the circular reflecting surface, we see that the reflected ray strikes the surface again at an angle of 65° to the normal and the above calculation applies. Thus a ray is deflected by 50° every time it strikes the mirror. The ray exits the cylinder once the total deflection angle exceeds 180°, which happens after 4 reflections.

7. Speed of electrons in an electrical wire is of the order of mm per second. The light goes on immediately when we close the switch because:

A. All electrons start moving and doing work at the same time after we close the switch

B. We are using AC voltage so electrons only have to move forth and back by a very short distance.

C. Emitting light from any bulb has nothing to do with movement of electrons.

D. Electric field, which forces the electrons to start moving, propagates with speed close to the speed of light.

E. A and B are true

F. A and D are true

Solution: F.
The light turns on immediately because the electric field propagates through the wire at nearly the speed of light, causing electrons throughout the circuit to start drifting almost simultaneously. This rapid response to the field powers the bulb, even though individual electrons move very slowly. Thus both statements A and D are true.

8. A person is floating on an inflatable mattress in the pool and drinking water from a metal cup. At some point this person drops the cup into the pool and the cup sinks. The level of the water in the pool:

A. stays the same

B. goes up

C. goes down

D. goes up or down depending how much water is in the cup.

Solution: C.
The difference in the volume of water displaced in each case is due to the metal cup only as all the other parameters (such as weight of the person and the volume of water in the cup) stay the same. The volume displaced by the cup is m_{cup}/ρ_w while it is being held by the person, whereas the volume displaced is $m_{\text{cup}}/\rho_{\text{cup}}$ when it sinks. As density of metal is higher than density of water the cup will displace less water than the volume of the water.

9. Two identical truck tires are deformed by the same amount: one very quickly when the car hits a curb, and the other very slowly when the truck bed is filled with snow. The ambient temperature is $0°C$ in both cases. Which deformation required a larger force to occur?

A. The fast one

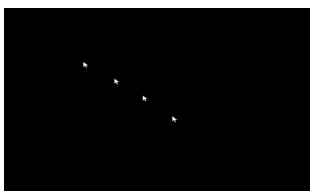
B. The slow one

C. Both needed the same force

D. Not enough information to decide

Solution: A.
The fast deformation is an adiabatic process with increasing temparature, whereas the slow deformation is an isothermal process. The final volume is the same in each case. Thus the fast process needs more work.

10. On the blank computer screen 18 cm high, the following picture of the cursor was observed when the mouse was moved. What was the speed of the mouse? The typical computer display works with 60 frames per second.



A. About 1 m/s
B. About 2 m/s
C. About 4 m/s
D. About 1 cm/s
E. About 2 cm/s
F. About 4 cm/s

Solution: B.
One can measure on the picture the height of the screen and the distance between the images of the pointer. Knowing that the height of the screen is $18cm$ one can calculate that distance between the images of the pointer is about 4 cm and this distance is covered in $1/60s$. This gives us the speed of about $2m/s$.

11. A microwave oven works at the frequency of 2.4 GHz. A large sheet of chocolate was placed in the working microwave (without the rotating plate). After it was removed, there was a distinct pattern of partly melted chocolate. The areas of melted chocolate were separated by area of non melted ones. What was the distance between the melted regions?

A. About 6 cm
B. About 12 cm
C. About 24 cm
D. About 48 cm

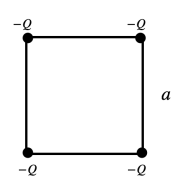
Solution: A.
Wavelength is about $3 \times 10^6/2.4 \times 10^7 = 12cm$. The distance between nodes of standing wave is half of the wavelength so about $6cm$.

12. One day when the temperature is $20^\circ C$ and pressure is close to the standard pressure, a hot air balloon with the air heated to $70^\circ C$ and volume of 1000 m^3 can lift about (including its own weight):

A. 1200 kg
B. 1020 kg
C. 170 kg
D. 340 kg
E. 860 kg

Solution: C.
Density of gas at constant pressure is inversely proportional to temperature in Kelvin. So the density of air at $70°C$ will be $1.2 \times 293/343 = 1.03$. Buoyant force will be about $1200 - 1030 = 170kg$.

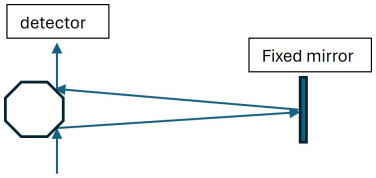
13. Four equal charges of $-Q$ are placed at the four corners of a square with side length a. At which point is the magnitude of the electric field the greatest?



A. In the centre of the square
B. In the middle of each side
C. Close to one of the corners
D. Far from the charges

Solution: C.
Using inverse square law, we see the magnitude of the field is the largest close to each charge.

14. A beam of light is directed towards an 8-sided rotating mirror. After getting reflected by one of the sides, light travels to a fixed mirror 60 km away, and reflects back to the rotating mirror. What is the minimum frequency of rotation of the mirror at which the light beam reflects along the original direction and reaches the detector?



A. 300000 rpm

B. 75000 rpm

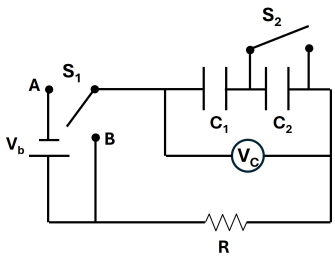
C. 37500 rpm

D. 18750 rpm

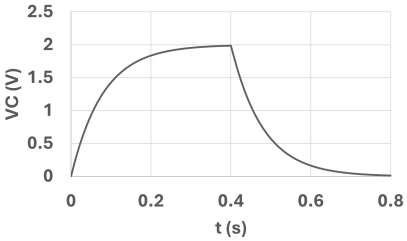
Solution: D.
The beam of light will be reflected along the original direction only if mirror rotates by 1/8 of the period during which light travels forth and back to the fixed mirror which is a distance L from the rotating mirror.

$$\frac{1}{8}T = \frac{2L}{c} \quad T = \frac{8 \times 2L}{c}$$
$$v = \frac{60 \times c}{8 \times 2L}\text{rpm}$$
$$v = \frac{60 \times 3 \times 10^8}{8 \times 2 \times 60 \times 10^3}\text{rpm} = 18750\text{rpm}$$

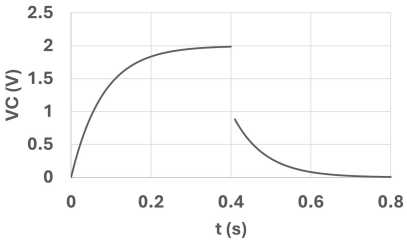
15. An RC circuit consists of two identical capacitors C_1 and C_2 , in series, a resistor R, a battery with voltage V_b , and two switches S_1 and S_2 initially open, as shown in the diagram. Initially switch S1 is set to position "B" and then at time 0 the it is moved to position A. At time 0.4 s switch S_1 is moved back to position B and at the same time the switch S_2 is closed. Which of the following graphs correctly represents the measured voltage as a function of time?



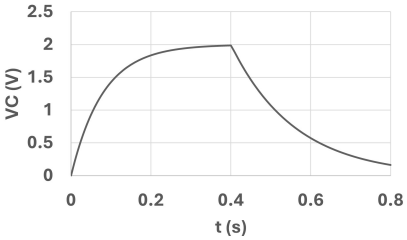
A.



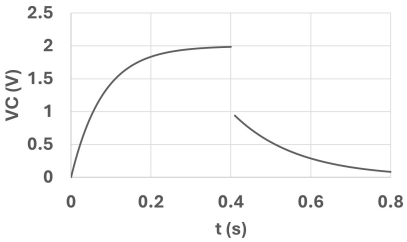
B.



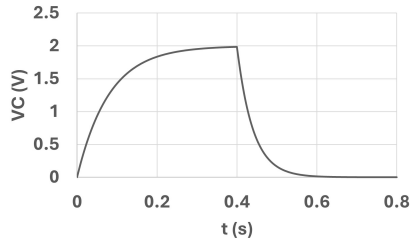
C.



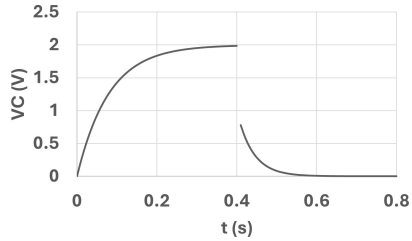
D.



E.

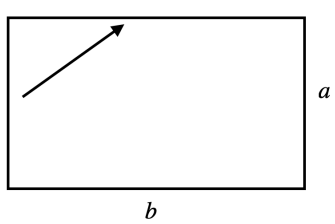


F.



Solution: D.
Closing switch S_2 discharges capacitor C_2. With only C_1 remaining in the circuit. The equivalent capacitance decreases, and as a result the time constant increases. Additionally, since $V_c = V_1 + V_2$ and now $V_2 = 0$ and V_1 are unchanged, V_c will drop.

16. The figure below shows the cross-section of a rectangular mirror with dimensions a and b. A light ray reflects between the mirrors such that its trajectory traces out a single rectangle with dimensions c and d. What is the ratio $\frac{a}{b}$?



A. $\frac{c}{d}$
B. $\frac{c^2d^2}{c^2+d^2}$
C. 1
D. $\frac{c}{\sqrt{d}}$
E. $\frac{\sqrt{d}}{c}$

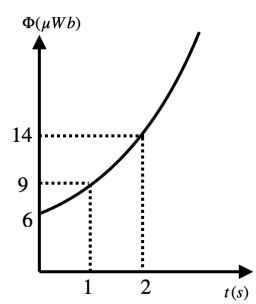
Solution: C.
$2i = 90$, $i = 45°$, $90 - i = 45$, $a = c\sin(45) + dsin(45)$, $b = c\cos(45) + d\cos(45)$ so $a/b = 1$

17. You are showing the trick of pulling a sheet of paper from under a glass of water without moving the glass much or spilling the water.

A. You should pull the paper fast to reduce the friction force between the paper and glass.
B. You have to pull paper very slowly to keep the glass from moving.
C. You are pulling the paper fast so the force of friction acts on the glass for a very short time.
D. The glass does not move at all if you are fast enough.

Solution: The friction force between the glass and the paper is independent of the speed at which you pull the paper. By pulling the paper fast you reduce the time over which this force acts, ensuring that the glass moves very little. Thus C is true.

18. The graph of the magnetic flux passing through a loop as a function of time is a parabolic curve. The axis of symmetry of the parabola is parallel to the ϕ axis. What is the magnitude of the induced electromotive force (emf) at the moment $t = 0$?



A. 0 μV
B. 1 μV
C. 2 μV
D. 3 μV

Solution: C.
Since Φ is a quadratic function of time, it must be of the form $\Phi = at^2 + bt + c$. From the two data points in the graph, we deduce that $\Phi = t^2+2t+6$. Taking a time derivative $d\Phi/dt$ gives us magnitude of the emf.

19. In old times, to start a delayed explosion, for example in mining, one would set fire to one end of a rope, which would burn slowly along its length, with the other end at the explosive material. Suppose you have two non-identical ropes, each of which takes one hour to burn its entire length. The ropes have varying thicknesses, so different sections burn at different rates. What is the minimum time interval you can measure with the two ropes?

    A. 1 hour

    B. 45 minutes

    C. 30 minutes

    D. 15 minutes

Solution: D.
The key is to realize that if a rope is set on fire from both ends, it takes 30 minutes to burn each rope (this is true irrespective of the nature of the cross-sections of the ropes). So, this is how one can measure 15 minutes with the system. One rope is set on fire from both ends and the other rope is set on fire from one end. After the first rope is completely burnt, it means 30 minutes have passed. At this instant, the second rope is to be set on fire from the other end. The time it takes for the second rope to burn from this instant is 15 minutes.

20. A Geiger-Muller counter is used to detect particles created during a radioactive atom's decay. The dead time of a Geiger-Muller counter is the interval between two detections within which the counter is unresponsive. That is right after a detection, the counter does not register a particle if it arrives within the dead time. Consider a counter recording $10,000$ counts per second. If the resolving time for the counter is 2×10^{-5} s, what is the best estimate of the actual count per second taking into account that the decay of a radioactive atom is a probabilistic event?

    A. $9,000 - 10,000$

    B. $10,000 - 11,000$

    C. $11,000 - 12,000$

    D. $12,000 - 13,000$

    E. It can be any random number.

Solution: None of the above options give the correct answer.
Denote the dead time by τ. If the counter records $N = 10000$ counts per second, then it was inactive for $N\tau$ seconds within the 1-second duration. Therefore, in $1 - N\tau$ seconds it registered N counts which implies it should record $n = N/(1 - N\tau) = 12500$ counts per second taking into account the dead time. Since the radioactive decay is a probabilistic event, the actual count can be any number between $10000 - 12500$.

21. A horizontal string is attached to an oscillator at one end and a hanging mass at the other. The oscillator's frequency is adjusted such that the string is vibrating in the second harmonic mode, forming a standing wave. Now, imagine immersing the hanging mass in water. Which of the following options is correct?

    A. The second harmonic mode will no longer be excited. It can be restored by increasing the oscillator's frequency.

    B. The second harmonic mode will no longer be excited. It can be restored by decreasing the oscillator's frequency.

    C. The second harmonic mode will no longer be excited. It cannot be restored by adjusting the frequency.

    D. The second harmonic mode will not be affected.

Solution: B.
Immersing the mass in water will reduce the tension in the string and this in turn will decrease the speed of the wave in the string. Standing wave can be restored by decreasing the frequency of the oscillator.

22. A horizontal string is attached to an oscillator that oscillates vertically at one end, and a hanging object with uniform density at the other. The

oscillator's frequency is adjusted such that the string is vibrating in the second harmonic mode, forming a standing wave. You immerse the hanging mass in water and observe that the string is now vibrating in the fourth harmonic mode. What is the density of the hanging mass?

A. 4/3 of the density of water

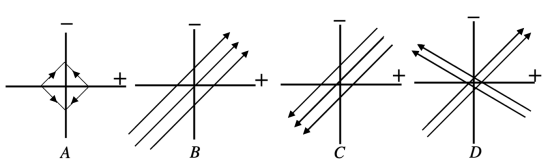
B. 2 times of the density of water

C. 3/4 of the density of water

D. 1/2 of the density of water

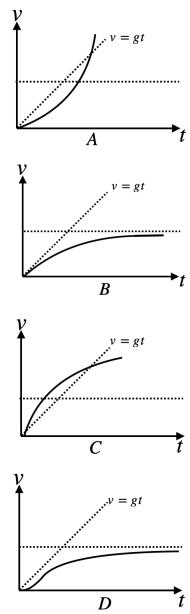
Solution: A.
The fact that the string is oscillating in the 4th harmonic after the mass is immersed in water suggests that the tension in the string is dropped to a quarter of the initial value, T=mg. This means the buoyant force on the mass must be $\frac{3}{4}$ mg: $F_b = \frac{3}{4}mg \rightarrow \rho_{\text{water}}\, Vg = \frac{3}{4}\rho_{\text{object}}\, Vg \rightarrow \rho_{\text{object}} = \frac{4}{3}\rho_{\text{water}}$

23. We know that the electric field generated by a single infinite flat sheet with a uniform charge distribution has a constant magnitude and is perpendicular to the surface. Now, consider two infinite non-conducting flat sheets with uniform but opposite charge distributions, perpendicular to each other. Which figure correctly shows the shape of the electric field lines between the sheets?



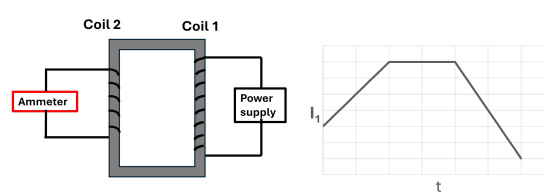
Solution: A.
Let E_0 be the constant magnitude of electric field produced by either of the charged sheets. In the first quadrant, the horizontal sheet produces an electric field $E_0\hat{y}$ whereas the vertical sheet produces a field $-E_0\hat{x}$, which when superposed give an electric field at an angle of $3\pi/4$ with the $x-$axis. Following a similar logic in the other quadrants we get a field configuration shown in option A.

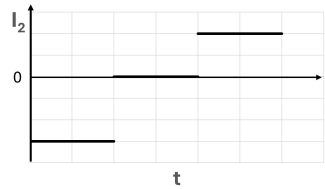
24. A raindrop begins to fall from rest at time $t = 0$ in the presence of resistive force $F = -\gamma v$, where v is the speed of the raindrop and γ is a constant. Which graph represents the velocity of the drop over time?

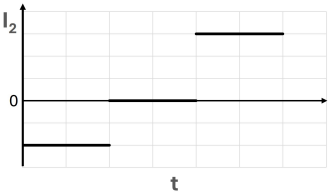


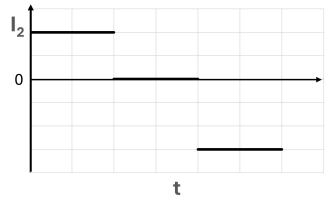
Solution: B.
As $v = 0$ initially, the only force acting on the raindrop is gravity, giving it an acceleration of g. After a very long time, the drop reaches terminal velocity and its net acceleration is zero. Thus the $v - t$ graph should have a positive slope g near $t = 0$, with the slope decreasing monotonically, approaching zero after a long time. Option B is consistent with the above description.

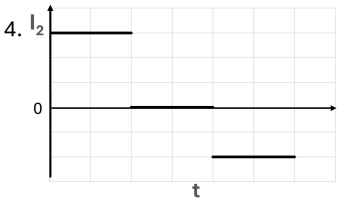
25. The following diagram shows two coils of a conducting wire wrapped around a core. The electric current running through coil 1 changes with time as shown in the picture on the right diagram. Which graph correctly describes the induced current in coil 2?



1. 

2. 

3. 

4. 

A. 1 only

B. 2 only

C. 3 only

D. 4 only

E. 1 or 4

F. 2 or 3

Solution: E.
According to Faraday's law and Lens's law, the current I_1 induced in the inner solenoid is proportional to the negative of the slope of I_2. Both figures 1 and 4 are consistent with this explanation. They differ only in the sign of I_2 which in turn depends on the direction of current I_1, and can not be deduced based on the information given in the question.

Section B

1. Two eggs, one hard boiled, the other raw were rotating for a long time at 20 rotations per second. At some point the torque forcing them to rotate was removed. Which one will stop first? Which one will be warmer after it stops? Estimate how much warmer. This will be just an order of magnitude calculation -you are not expected to give an accurate value, just an estimate. The rotational kinetic energy of the sphere is $\frac{1}{2}I\omega^2$, where I is the moment of inertia of the sphere about the center of mass, and ω is the angular speed. The moment of inertia of a solid sphere is given by the formula $I = \frac{2}{5}MR^2$, where R is the radius and M is the mass of the sphere.

Solution: All the kinetic energy of the egg has to be converted to heat when the egg stops. The stopping of the hard boiled egg is mainly due to air resistance and friction between the egg and the surface, on which it rotates, so only fraction of it is dissipated in the egg. For the raw egg faster stopping is due to the drag of the liquid sloshing inside, so more of the kinetic energy heats the inside of the egg. So the raw egg stops first and is warmer after stopping. Assuming that more of the kinetic energy goes into heating the raw egg, and assuming that the egg is a ball of radius 15 mm filled with water we get:

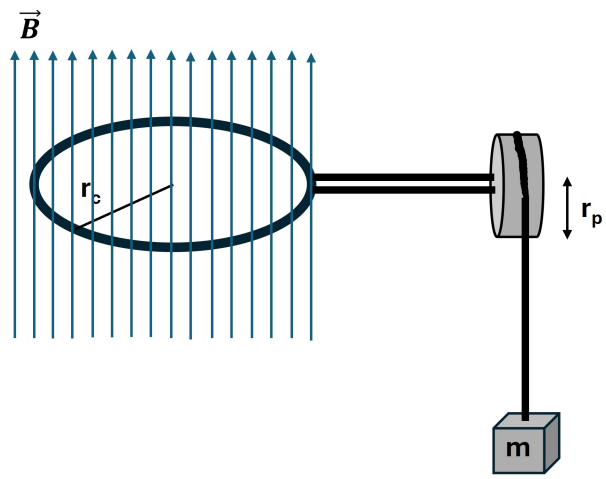
$$mC_w\Delta T = \frac{1}{2}I\omega^2 = \frac{1}{2} \times \frac{2}{5}mR^2 \times \omega^2$$

$$\Delta T = 1/5 \times 0.0152 \times (2\pi \times 20)/4186 = 0.00017 \sim 0.2\ mK$$

2. A circular copper coil consisting of 90 loops of radius $r_c = 0.35m$, and a total resistance $R = 4.2\Omega$,is attached to a plastic pulley with radius $r_p = 0.4m$, as shown in the figure below. A non-elastic rope of negligible mass is wrapped around the pulley and attached to a hanging mass $m = 2.0kg$. When released, the rope unwinds without slipping as the hanging mass drops. A uniform magnetic field $B = 0.1T$ is initially perpendicular to the plane of the coil.

   a. Derive an expression and calculate the numerical value of the average terminal speed (the terminal speed might vary slightly during each revolution of the coil).

   b. Sketch a graph of velocity of the hanging mass as a function of time. You don't need to include any numerical values.



Solution: Because of the oscillatory nature of the force acting on the loop, there will be never a constant terminal velocity - it will always oscillate around some average constant value. This average will be achieved when the power lost as heat in the resistance of the coil V^2/R will be equal to the power of the gravitational force of the weight (mgv).

The voltage induced in the coil is:

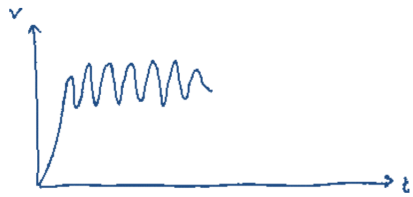
$$V = N\frac{d}{dt}(\boldsymbol{A}\cdot\boldsymbol{B}) = N\frac{d}{dt}\left(\pi r_c^2 B\cos\omega t\right) = N\frac{d}{dt}\left(\pi r_c^2 B\cos\frac{vt}{r_p}\right) = N\pi r_c^2 B\frac{v}{r_p}\sin\frac{vt}{r_p}$$

Average over a revolution gives:

$$V^2 = \frac{1}{2}\left(N\pi r_c^2 B\frac{v}{r_p}\right)^2$$

Setting $V^2/R = mgv$, we get

$$1/2(N\pi r_c^2 Bv/r_p)^2 = mgvR \quad \Longrightarrow \quad v = \frac{2mgRr_p^2}{(N\pi r_c^2 B)^2} = 2.21m/s$$



Alternatively;

For the hanging mass $mg - T = ma$. For the pulley, $Tr_p - \tau_{\text{counter}} = \frac{I_{\text{pulley}}a}{r_p}$. This gives us

$$\tau_{\text{counter}} = \boldsymbol{\mu} \times \mathbf{B} = NiAB\sin\theta$$

$$i = \frac{d\phi/dt}{R} = \frac{1}{R}\frac{d(N\mathbf{A}.\mathbf{B})}{dt} = \frac{N\pi r_c^2 B\sin\theta}{R}\frac{d\theta}{dt}$$

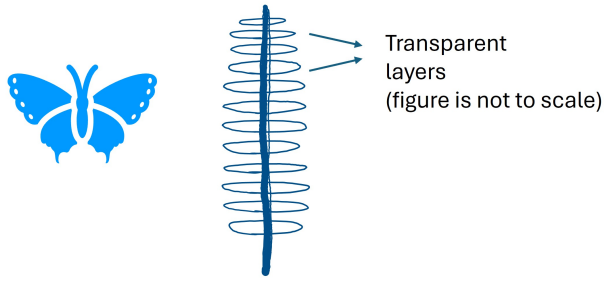
$$\Longrightarrow \tau_{\text{counter}} = \frac{(N\pi r_c^2 B\sin\theta)^2}{R}\frac{v}{r_p}$$

When approaching the terminal speed $T = mg$ and $\tau_{\text{counter}} = Tr_p$. This gives us

$$\frac{(N\pi r_c^2 B\sin\theta)^2}{R}\frac{v}{r_p} = mgr_p.$$

Solving the above equation for v and using that average of $\sin^2\theta$ over a period is $1/2$, we get $v = \frac{2mgRr_p^2}{(N\pi r_c^2 B)^2} = 2.21m/s$.

3. The blue color of Morpho butterfly's wings is not due to pigment molecules but is instead caused by the light interference. This interference occurs when light reflects off the nano-size, tree-shaped structure on the wing's surface called lamellae (See the picture below). These lamellae consist of thin, transparent layers ($54nm$ thick) with a refractive index of 1.56, separated by air gaps.



a. Considering the perpendicular reflections from only the top two transparent layers, calculate the size of the air gap between the layers. ($\lambda_{\text{blue}} = 452nm$)

b. Calculate the wavelength of the light reflected by the wing if it were dipped in liquid nitrogen. Assume that liquid nitrogen would replace the air gaps and would not change the dimensions of the wing. The index of refraction of liquid nitrogen is 1.20.

c. As the butterfly flies, an observer will see the reflections of light incident on the wings at different angles. Considering the reflection from only the top two layers, calculate the wavelengths of the observed light for the angles of incidence of $5°, 10°, 20°, 30°, 40°$, and $45°$? What is the color of the observed light during flight?

Solution:

a)

$$2\left(n_{\text{layer}}\, d_{\text{layer}} + n_{\text{air}}\, d_{\text{air}}\right) = \lambda \rightarrow \quad d_{\text{air}} = \frac{\frac{\lambda}{2} - n_{\text{layer}}\, d_{\text{layer}}}{n_{\text{air}}} = \frac{\frac{452}{2} - 1.56 \times 54}{1} = 142 \text{ nm}$$

b) The air gap would be filled with liquid nitrogen:

$$\lambda = 2\left(n_{\text{layer}}\, d_{\text{layer}} + n_{\text{nitrogen}}\, d_{\text{nitrogen}}\right) = 2(1.56 \times 54 + 1.20 \times 142) = 509 \text{ nm}$$

The wing would appear green.

c) The wavelength λ is given by:

$$\lambda = 2\,(n_{\text{layer}}\ d_{\text{layer}}\ \cos\theta_{\text{layer}}\ + n_{\text{air}}\ d_{\text{air}}\ \cos\theta_{\text{air}}\ ) = 2\,(1.56 \times 54 \times + \cos\theta_{\text{layer}}\ + 142\cos\theta_{\text{air}}\ )$$

Snell's law relates the angles:

$$n_{\text{layer}} \sin\theta_{\text{layer}} = n_{\text{air}} \sin\theta_{\text{air}} \quad \Rightarrow \quad \cos\theta_{\text{layer}} = \sqrt{1 - \left(\frac{n_{\text{air}}}{n_{\text{layer}}} \sin\theta_{\text{air}}\right)^2}$$

The following data points show the variation of the perceived color with the incidence angle:

θ_{air}	$\cos\theta_{\text{air}}$	$\cos\theta_{\text{layer}}$	λ (nm)
5°	0.9962	0.9984	451
10°	0.9848	0.9938	447
20°	0.9397	0.9757	431
30°	0.8660	0.9472	405
40°	0.7660	0.9112	371
45°	0.7071	0.8914	351

As the angle increases, the reflected light shifts toward shorter wavelengths, resulting in darker blue, violet, and eventually ultraviolet hues.

Question 1	●	b	c	d	e	f
Question 2	a	b	c	●	e	f
Question 3	a	●	c	d	e	f
Question 4	a	b	●	d	e	f
Question 5	a	●	c	d	e	f
Question 6	a	b	c	●	e	f
Question 7	a	b	c	d	e	●
Question 8	a	●	c	d	e	f
Question 9	●	b	c	d	e	f
Question 10	a	●	c	d	e	f
Question 11	●	b	c	d	e	f
Question 12	a	b	●	d	e	f
Question 13	a	b	●	d	e	f
Question 14	a	b	c	●	e	f
Question 15	a	b	c	●	e	f
Question 16	a	b	●	d	e	f
Question 17	a	b	●	d	e	f
Question 18	a	b	●	d	e	f
Question 19	a	b	c	●	e	f
Question dropped						
Question 21	a	●	c	d	e	f
Question 22	●	b	c	d	e	f
Question 23	●	b	c	d	e	f
Question 24	a	●	c	d	e	f
Question 25	a	b	c	d	●	f